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In [1]: import torch
import torch.nn as nn
from torch import optim
import matplotlib.pyplot as plt
from IPython import display
from d2l import torch as d2l
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In [2]: def train(net, train_iter, test_iter, loss, num_epochs, updater, GPU = False):
        """训练模型（使用GPU）"""
        # 检查并设置设备
        if GPU == True:
            device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
        else:
            device = torch.device('cpu')
        print(f'Using device: {device}')

        # 将模型移动到GPU
        net = net.to(device)

        animator = Animator(xlabel='epoch', xlim=[1, num_epochs], ylim=[0.3, 0.9],
                            legend=['train loss', 'train acc', 'test acc'])

        for epoch in range(num_epochs):
            train_metrics = train_epoch_gpu(net, train_iter, loss, updater, device)
            test_acc = evaluate_accuracy_gpu(net, test_iter, device)
            animator.add(epoch + 1, train_metrics + (test_acc,))
        train_loss, train_acc = train_metrics
        print(f'final train loss {train_loss:.3f}, '
              f'train acc {train_acc:.3f}, test acc {test_acc:.3f}')
        # assert train_loss < 0.5, train_loss
        # assert train_acc <= 1 and train_acc > 0.7, train_acc
        # assert test_acc <= 1 and test_acc > 0.7, test_acc

    def train_epoch_gpu(net, train_iter, loss, updater, device):
        """训练模型一个迭代周期（使用GPU）"""
        # 设置模型为训练模式
        if isinstance(net, torch.nn.Module):
            net.train()

        metric = Accumulator(3)

        for X, y in train_iter:
            # 将数据移动到GPU
            X = X.to(device)
            y = y.to(device)

            # 前向传播
            y_hat = net(X)
            l = loss(y_hat, y)

            # 反向传播
            if isinstance(updater, torch.optim.Optimizer):
                updater.zero_grad()
                l.backward()
                updater.step()
            else:
                l.sum().backward()
                updater(X.shape[0])
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        # 记录指标
        with torch.no_grad():
            metric.add(float(1.sum()), accuracy(y_hat, y), y.numel())

# 返回训练损失和训练精度
return metric[0] / metric[2], metric[1] / metric[2]

def evaluate_accuracy_gpu(net, data_iter, device):
    """计算模型在指定数据集上的精度（使用GPU）"""
    if isinstance(net, torch.nn.Module):
        net.eval() # 设置为评估模式

    metric = Accumulator(2)

    with torch.no_grad():
        for X, y in data_iter:
            # 将数据移动到GPU
            X = X.to(device)
            y = y.to(device)

            # 计算精度
            metric.add(accuracy(net(X), y), y.numel())

    return metric[0] / metric[1]

def accuracy(y_hat, y):
    """计算预测正确的数量"""
    if len(y_hat.shape) > 1 and y_hat.shape[1] > 1:
        y_hat = y_hat.argmax(axis=1)
    cmp = y_hat.type(y.dtype) == y
    return float(cmp.type(y.dtype).sum())

class Accumulator:
    """在n个变量上累加"""
    def __init__(self, n):
        self.data = [0.0] * n

    def add(self, *args):
        self.data = [a + float(b) for a, b in zip(self.data, args)]

    def reset(self):
        self.data = [0.0] * len(self.data)

    def __getitem__(self, idx):
        return self.data[idx]

class Animator:
    """在动画中绘制数据"""
    def __init__(self, xlabel=None, ylabel=None, legend=None, xlim=None,
                 ylim=None, xscale='linear', yscale='linear',
                 fmts=('-', 'm--', 'g-.', 'r:'), nrows=1, ncols=1,
                 figsize=(3.5, 2.5)):
        # 增量地绘制多条线
        if legend is None:

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        legend = []
    d2l.use_svg_display()
    self.fig, self.axes = d2l.plt.subplots(nrows, ncols, figsize=figsize)
    if nrows * ncols == 1:
        self.axes = [self.axes, ]
    # 使用Lambda函数捕获参数
    self.config_axes = lambda: d2l.set_axes(
        self.axes[0], xlabel, ylabel, xlim, ylim, xscale, yscale, legend)
    self.X, self.Y, self.fmts = None, None, fmts

def add(self, x, y):
    # 向图表中添加多个数据点
    if not hasattr(y, "__len__"):
        y = [y]
    n = len(y)
    if not hasattr(x, "__len__"):
        x = [x] * n
    if not self.X:
        self.X = [[] for _ in range(n)]
    if not self.Y:
        self.Y = [[] for _ in range(n)]
    for i, (a, b) in enumerate(zip(x, y)):
        if a is not None and b is not None:
            self.X[i].append(a)
            self.Y[i].append(b)
    self.axes[0].cla()
    for x, y, fmt in zip(self.X, self.Y, self.fmts):
        self.axes[0].plot(x, y, fmt)
    self.config_axes()
    display.display(self.fig)
    display.clear_output(wait=True)

class Animator:
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        ylim=None, xscale='linear', yscale='linear',
        fmts=('-', 'm--', 'g-.', 'r:'), nrows=1, ncols=1,
        figsize=(3.5, 2.5)):
        # 增量地绘制多条线
        if legend is None:
            legend = []
        d2l.use_svg_display()
        self.fig, self.axes = d2l.plt.subplots(nrows, ncols, figsize=figsize)
        if nrows * ncols == 1:
            self.axes = [self.axes, ]
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    def add(self, x, y):
        # 向图表中添加多个数据点
        if not hasattr(y, "__len__"):
            y = [y]
        n = len(y)
        if not hasattr(x, "__len__"):
            x = [x] * n
        if not self.X:
            self.X = [[] for _ in range(n)]
        if not self.Y:
            self.Y = [[] for _ in range(n)]

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        for i, (a, b) in enumerate(zip(x, y)):
            if a is not None and b is not None:
                self.X[i].append(a)
                self.Y[i].append(b)
        self.axes[0].cla()
        for x, y, fmt in zip(self.X, self.Y, self.fmts):
            self.axes[0].plot(x, y, fmt)
        self.config_axes()
        display.display(self.fig)
        display.clear_output(wait=True)

def predict(net, test_iter, n=6):
    """预测标签"""
    for X, y in test_iter:
        break
    trues = d2l.get_fashion_mnist_labels(y)
    preds = d2l.get_fashion_mnist_labels(net(X).argmax(axis=1))
    titles = [true + '\n' + pred for true, pred in zip(trues, preds)]
    d2l.show_images(
        X[0:n].reshape((n, 28, 28)), 1, n, titles=titles[0:n])

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In [3]: # num_inputs, num_outputs, num_hiddens = 784, 10, 256

# W1 = nn.Parameter(torch.randn(
#     num_inputs, num_hiddens, requires_grad=True) * 0.01)
# b1 = nn.Parameter(torch.zeros(num_hiddens, requires_grad=True))
# W2 = nn.Parameter(torch.randn(
#     num_hiddens, num_outputs, requires_grad=True) * 0.01)
# b2 = nn.Parameter(torch.zeros(num_outputs, requires_grad=True))

# params = [W1, b1, W2, b2]

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In [4]: # def relu(X):
#     a = torch.zeros_like(X)
#     return torch.max(X, a)

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In [5]: # def net(X):
#     X = X.reshape((-1, num_inputs))
#     H = relu(X@W1 + b1) # 这里“@”代表矩阵乘法
#     return (H@W2 + b2)

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In [6]: batch_size, lr, num_epochs = 256, 0.3, 100
train_iter, test_iter = d2l.load_data_fashion_mnist(batch_size)
# 定义损失函数和优化器
loss = nn.CrossEntropyLoss()

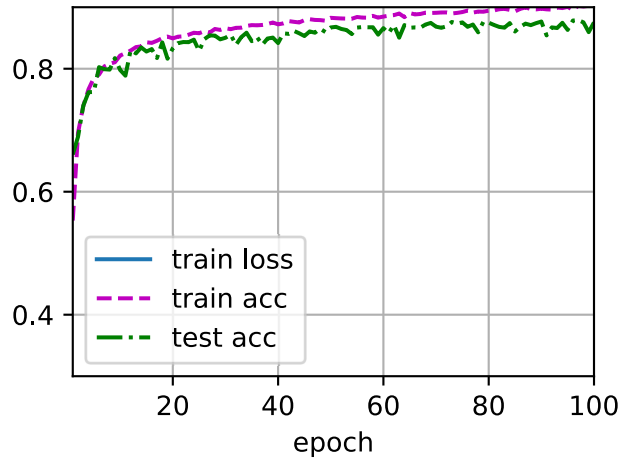
```

```

In [7]: %%time
net = nn.Sequential(
    nn.Flatten(), # 将28x28图像展平为784维向量
    nn.Linear(784, 256), # 全连接层: 784 → 256
    nn.ReLU(), # 激活函数
    nn.Linear(256, 10) # 输出层: 256 → 10 (对应10个类别)
)
optimizer = torch.optim.SGD(net.parameters(), lr=lr)
# 使用GPU训练
train(net, train_iter, test_iter, loss, num_epochs=num_epochs, updater=optimizer

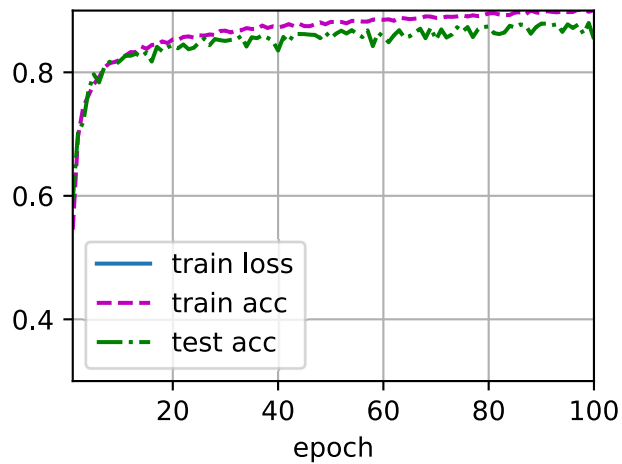
```

final train loss 0.000, train acc 0.900, test acc 0.875
 CPU times: total: 27.1 s
 Wall time: 10min 22s



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In [8]: %%time
net1 = nn.Sequential(nn.Flatten(),
                    nn.Linear(784, 256),
                    nn.ReLU(),
                    nn.Linear(256, 10))
optimizer = torch.optim.SGD(net1.parameters(), lr=lr)
train(net1, train_iter, test_iter, loss, num_epochs=num_epochs, updater=optimizer)
predict(net1, test_iter)
```

final train loss 0.000, train acc 0.901, test acc 0.854
 CPU times: total: 2min 19s
 Wall time: 10min 32s



```
In [9]: import torch
import subprocess

print("PyTorch版本:", torch.__version__)
```

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print("CUDA是否可用:", torch.cuda.is_available())
print("CUDA版本:", torch.version.cuda)

# 检查系统GPU信息
try:
    result = subprocess.run(['nvidia-smi'], capture_output=True, text=True)
    if result.returncode == 0:
        print("nvidia-smi输出:")
        print(result.stdout)
    else:
        print("nvidia-smi命令失败，可能没有NVIDIA显卡或驱动未安装")
except FileNotFoundError:
    print("nvidia-smi未找到，请安装NVIDIA驱动")
```

PyTorch版本: 2.9.0+cu130

CUDA是否可用: True

CUDA版本: 13.0

nvidia-smi输出:

Tue Nov 11 16:38:43 2025

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+-----+
+-----+
| NVIDIA-SMI 581.08                 Driver Version: 581.08          CUDA Version:
13.0   |
+-----+-----+-----+
+-----+
| GPU  Name                       Driver-Model | Bus-Id          Disp.A | Volatile Unc
orr. ECC |
| Fan  Temp   Perf           Pwr:Usage/Cap |      Memory-Usage | GPU-Util  Co
mpute M. |
|      |      |      |      |      |      |
| MIG M. |      |      |      |      |      |
|=====+=====+=====+
=====|
|   0   NVIDIA GeForce RTX 5070 Ti   WDDM | 00000000:02:00.0 On |
N/A |
|   0%   45C    P8              24W / 300W |   3443MiB / 16303MiB |      1%
Default |
|      |      |      |      |      |      |
| N/A |      |      |      |      |      |
+-----+-----+-----+
+-----+
+-----+
| Processes:
|
| GPU   GI    CI          PID    Type    Process name                        GP
U Memory |
|      ID    ID          |      |      |
|      |      |      |      |      |      |
|=====+=====+=====+
=====|
|   0   N/A   N/A         2152    C+G    ...ntrolPanel\SystemSettings.exe
N/A |
|   0   N/A   N/A         2460    C+G    ...0.3595.53\msedgewebview2.exe
N/A |
|   0   N/A   N/A         3408      C    ...s\Python\Python312\python.exe
N/A |
|   0   N/A   N/A         4372    C+G    ...c\AdobeNotificationClient.exe
N/A |
|   0   N/A   N/A         5688    C+G    ..._cw5n1h2txyewy\SearchHost.exe
N/A |
|   0   N/A   N/A         8596    C+G    ...lpaper_engine\wallpaper32.exe
N/A |
|   0   N/A   N/A         9840    C+G    ...lus\logioptionsplus_agent.exe
N/A |
|   0   N/A   N/A        11724    C+G    ...xyewy\ShellExperienceHost.exe
N/A |
|   0   N/A   N/A        12552    C+G    ...logioptionsplus_logivoice.exe
N/A |
|   0   N/A   N/A        13076    C+G    ...em32\ApplicationFrameHost.exe
N/A |
|   0   N/A   N/A        13600    C+G    C:\Windows\explorer.exe
N/A |
```

	0	N/A	N/A	14000	C+G	...indows\System32\ShellHost.exe
N/A						
	0	N/A	N/A	15548	C+G	...t\Edge\Application\msedge.exe
N/A						
	0	N/A	N/A	16648	C+G	...y\StartMenuExperienceHost.exe
N/A						
	0	N/A	N/A	18988	C+G	...mba6cd70vzyy\ArmouryCrate.exe
N/A						
	0	N/A	N/A	19132	C+G	...ogram Files\ToDesk\ToDesk.exe
N/A						
	0	N/A	N/A	19600	C+G	...ce\root\Office16\POWERPNT.EXE
N/A						
	0	N/A	N/A	19708	C+G	...App_cw5n1h2txyewy\LockApp.exe
N/A						
	0	N/A	N/A	20804	C+G	...0.3595.53\msedgewebview2.exe
N/A						
	0	N/A	N/A	21228	C+G	...8bbwe\PhoneExperienceHost.exe
N/A						
	0	N/A	N/A	22556	C+G	...0.3595.53\msedgewebview2.exe
N/A						
	0	N/A	N/A	24076	C+G	...5n1h2txyewy\TextInputHost.exe
N/A						
	0	N/A	N/A	26112	C+G	...crosoft OneDrive\OneDrive.exe
N/A						
	0	N/A	N/A	26316	C+G	...yb3d8bbwe\WindowsTerminal.exe
N/A						
	0	N/A	N/A	26896	C+G	...\Creative Cloud UI Helper.exe
N/A						
	0	N/A	N/A	28060	C+G	...Chrome\Application\chrome.exe
N/A						
	0	N/A	N/A	28184	C+G	...are\IntelGraphicsSoftware.exe
N/A						
	0	N/A	N/A	29752	C+G	...ef.win7x64\steamwebhelper.exe
N/A						
	0	N/A	N/A	31424	C+G	...Claude\app-1.0.332\claude.exe
N/A						
	0	N/A	N/A	32232	C+G	...26wp6bftszj\TranslucentTB.exe
N/A						
	0	N/A	N/A	32616	C+G	...8wekyb3d8bbwe\M365Copilot.exe
N/A						
	0	N/A	N/A	34508	C+G	...0.3595.53\msedgewebview2.exe
N/A						
	0	N/A	N/A	35316	C+G	...bin-7.2.10\NutstoreClient.exe
N/A						
	0	N/A	N/A	35832	C+G	...2.10\Nutstore.WindowsHook.exe
N/A						

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In []:

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